Trustworthy Collective Reasoning and Evaluation for AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

Problem. Large language models (LLMs) increasingly power teams of AI agents: one proposes an answer, others evaluate it, challenge its reasoning, or contribute missing evidence. Yet majority voting and LLM-as-a-Judge scores can be confidently wrong when agents repeat the same error, favor their own outputs, or converge through discussion without independent verification. Consider a naval logistics assistant reporting that a replacement component is available: several evaluator agents endorse its recommendation, but each cites the same stale inventory snapshot while an updated depot record tells a different story. How can a collective distinguish genuine corroboration from repeated mistakes and then correct its reasoning?

Central hypothesis. We hypothesize that collective reasoning becomes more reliable when agents jointly estimate peer expertise, account for dependent judgments, and resolve disagreements through targeted evidence checks rather than additional votes. The goal is not merely a better model ranking: a trustworthy collective should know which claims remain unsupported, improve an answer through critique, and abstain when its apparent consensus is unreliable.

Research plan.

- **Learn reliable peer evaluation.** Build on peer ranking to assess both candidate answers and intermediate reasoning steps. Record independent, blinded judgments before discussion; model task-dependent judge ability, question difficulty, self-preference, and shared-source effects. Infer relative reliability from unlabeled peer comparisons, then calibrate correctness and uncertainty using a small independent labeled set when available. Study when judge weights and dependence are identifiable, and when agreement cannot support an absolute confidence claim.
- **Turn disagreement into productive reasoning.** Develop structured deliberation in which agents identify the earliest contested inference, request a specific source, test a counterexample, or verify a tool result. Separate evidence-seeking critiques from persuasive but unsupported discussion, and track how each exchange changes the proposed answer and its support. We will test whether independent challenges rescue correct minority views while avoiding harmful revisions of initially correct answers.
- **Use collective feedback to improve agent decisions.** Learn which peers or checks to consult next based on complementary strengths, uncertainty, and the type of suspected failure. Feed verified corrections and process-level critiques back into agent training so evaluation improves future reasoning, not just a leaderboard. Investigate policies for revision, escalation, and abstention when evidence remains inadequate; computation and interaction cost will be measured, not treated as the research objective.

Evaluation and expected outcomes. Evaluate on MEQA, MuSiQue, and MATH, plus controlled multistep cases with duplicated sources, self-preference, and deliberately correlated errors. Compare with a single strong judge, majority voting, unmodified peer ranking, best-of-N, and unconstrained multi-agent debate. Report agreement with independent reference judgments, ranking stability, confidence calibration, final-answer accuracy after revision, recovery of correct minority answers, harmful answer flips, and evaluation cost. Hold out topics and agent families to test transfer.

Naval relevance. Naval analysts synthesizing maintenance, readiness, and mission-status reports may receive several AI assessments that appear independent but all trace to one outdated record. A collective reasoning assistant could expose the shared source, solicit a genuinely new check, and explain whether the evidence warrants an answer or escalation. This fits Neskovic's interests in large-scale evaluation of distributed information, collaborative problem solving, expertise assessment, and principled decision making under uncertainty.

Prior work and PI qualifications. Our PRD peer-ranking and discussion work (Li et al., 2024, TMLR) provides a foundation for aggregating model judgments; IQA-EVAL (Li et al., 2024, NeurIPS) studies interactive evaluation; and FG-PRM (Li et al., 2025, Findings of EMNLP) provides fine-grained feedback on reasoning errors. The PI holds a 2024 NSF CAREER award and received a 2023 Amazon Research Award for process-guided fine-tuning. As a postdoc, the PI led system integration for the RESIN team's RESIN-11 delivery system in DARPA KAIROS.